

Robotics and Human-Robot Interaction

Exam-Friendly Complete Summary

Prepared from: Intelligent Robotics and Automation lecture slides and Chapter 2, "What Is Human-Robot Interaction?"

Goal of this PDF: help you revise every slide/topic quickly, understand the main ideas, and write clear exam answers. It includes complete notes, slide-by-slide checklists, comparison tables, common mistakes, short questions, and model answers.

Chapter/Source	What you must master
Robotics lecture	Robot types, history, jobs and automation, SPA paradigm, Asimov laws, mobile/autonomous robots, humanoids, service/social/surgical/rescue robots, intelligence and AI.
HRI chapter	Meaning of HRI, physical vs social interaction, embodiment, HRI vs HCI, interdisciplinary nature, barriers, evolution of social robots, landmark robots, and ethical/social questions.
Exam strategy	Define first, compare clearly, give examples from slides, then explain why the topic matters for robot design and human use.

Recommended study order: 1) read the Quick Revision Map, 2) study Chapter 1 notes, 3) use the slide-by-slide checklist, 4) study Chapter 2 notes, 5) answer the model questions without looking.

Source note: This guide is a paraphrased study aid. It does not replace the original lecture slides or textbook chapter.

0. Quick Revision Map

This page is your last-hour overview. Learn the logic first, then memorize examples.

Big idea	Simple meaning	Remember this for exam
Robot	A machine that senses, processes information, and acts in the world.	Use the sense -> plan -> act logic when explaining any robot.
Robotics	The study and development of robots, including hardware, software, sensing, control, and applications.	Robotics is not only AI; it also includes mechanical design, electronics, sensors, actuators, and control.
Industrial robot	Robot used in factories and production.	Examples: welding, assembly, painting, material handling. Strength: precision, speed, repeatability.
Service robot	Robot that performs useful services outside classic industrial production.	Examples: cleaning robot, hospital robot, delivery robot, domestic assistant.
Social robot	Robot designed to interact socially with people.	Examples: Paro, Pepper, Nao, Kismet. Key idea: social cues and human expectations.
Mobile robot	Robot that moves in its environment with some autonomy.	Examples: rover, autonomous car, drone, AUV. Must navigate and avoid obstacles.
SPA paradigm	Sense -> model/plan -> control -> execute.	Classic robot control pipeline; good for structured tasks but can be slow in dynamic environments.
HRI	Study/design of interaction between humans and robots.	Unique because robots are physically embodied and people treat them as social actors.
Embodiment	Robot has a body that affects how it senses, acts, and is interpreted by people.	Eyes, mouth, size, shape, motion, and distance influence user expectations.
AI	Computer systems that model/apply human-like intelligence.	Intelligence: learn, solve novel problems, act rationally, act like humans.

Exam focus: Do not memorize only names. For each robot/example, remember: type, task, why it matters, and one limitation.

How to write any good answer

Step	What to write	Example
1. Definition	Start with a direct definition.	A mobile robot is a robot that moves relative to its environment with some degree of autonomy.
2. Components	Mention sensors, processing/control, and actuators.	A drone uses sensors and onboard control to fly and perform tasks.
3. Example	Use a slide example.	Example: Curiosity rover, Google car, Atlas, Paro, Nao.
4. Importance	Explain why it matters.	It improves automation, safety, service delivery, exploration, or social support.
5. Limitation/et hics	Mention safety, reliability, cost, jobs, privacy, or user trust.	A social robot can help users, but it must avoid over-trust and privacy problems.

1. Chapter 1 - Robotics Lecture: Complete Exam Notes

This section covers the topics shown across the robotics lecture slides, including history, robot categories, automation, robot examples, intelligence, and AI.

1.1 What is robotics?

- Robotics is the interdisciplinary field concerned with designing, building, programming, controlling, and applying robots.
- A robot usually has three core abilities: it can sense the environment, decide/process information, and act using actuators such as motors, wheels, arms, grippers, speakers, or screens.
- Robotics connects computer science, AI, mechanical engineering, electrical engineering, control systems, design, and human factors.
- The opening slides use evolution/history visuals to show that robots developed from human imagination, myths, automata, industrial machines, and finally AI-enabled systems.

Exam focus: A safe exam definition: "Robotics is the study and application of robots that sense their environment, process information, and act physically to perform tasks with some level of autonomy."

1.2 Robotics history and AI timeline

The history slides are mostly visual, but the message is clear: robotics did not appear suddenly. It grew through imagination, mechanical invention, automation, computing, and AI.

Stage	What happened	Why it matters
Ancient imagination	Myths and stories described artificial beings and mechanical helpers.	Shows that the dream of artificial life is very old.
Automata and mechanical devices	Clockwork machines and designed mechanisms imitated motion.	Started the idea that machines can reproduce actions.
Industrial automation	Robots began to perform repetitive factory tasks.	Robots became practical tools for productivity.
Computing and AI	Robots gained sensors, software, planning, and intelligent behavior.	Enabled autonomous vehicles, drones, service robots, and social robots.
Modern robotics	Robots now appear in homes, hospitals, factories, agriculture, space, underwater, and society.	Modern robotics is both technical and social.

Exam focus: History question answer pattern: "Robotics evolved from fictional/mechanical ideas to industrial automation and then to AI-enabled autonomous and social systems."

1.3 Robot market and application areas

The spending chart in the lecture groups robotics into military, industrial, commercial, and personal areas. The central lesson is that robotics has moved beyond factories into many parts of life.

Area	Examples	Main value	Risk/concern
Military	UAVs, UGVs, underwater vehicles, task robots	Remote operation, surveillance, dangerous missions	Ethics, safety, weaponization
Industrial	Welding, assembly, material handling, painting	Speed, precision, productivity, repeatability	Job displacement, safety near workers
Commercial/service	Medical robots, agricultural robots, construction robots, delivery robots	Improved services and efficiency	Cost, reliability, regulation
Personal/domestic	Cleaning robots, education robots, entertainment robots, home assistants	Convenience and accessibility	Privacy, over-dependence, user trust

1.4 Types of robots from the lecture

Type	Easy definition	Examples in slides	Exam notes
Industrial robots	Robots used in manufacturing and production.	Car manufacturing arms, welding robots, factory arms.	Usually fixed or semi-fixed, strong, precise, repetitive, and designed for productivity.
Service robots	Robots that help people by performing useful tasks outside classic factories.	Hospital/service robots, Roomba, assistive robots.	Can operate in homes, hospitals, offices, and public spaces.
Social robots	Robots made to interact with people through social cues.	Paro, Pepper-like robots, child/eldercare robots.	Design must consider speech, gestures, emotion, trust, and expectations.
Mobile robots	Robots that move relative to their environment with some autonomy.	Rovers, autonomous cars, drones, legged robots.	Need navigation, localization, obstacle avoidance, and energy management.
Humanoid robots	Robots shaped like the human body or parts of it.	ASIMO, Nao, Geminoid, CB2, Atlas.	Useful for human environments but difficult to control and expensive.
Autonomous vehicles	Vehicles that sense and move with limited or no human control.	Google car, Urban Challenge vehicles, UAV/UGV/AUV.	Require perception, planning, safety, mapping, and decision-making.
Surgical robots	Robots that assist in medical/surgical procedures.	Da Vinci-type robotic systems.	Improve precision, but require trained operators and safety controls.
Rescue robots	Robots used in disasters or dangerous environments.	Search-and-rescue robots.	Protect humans by entering risky places.

1.5 Jobs and automation

The lecture asks what jobs could be taken over by robots and shows examples such as chef, retail sales associate, farmer, factory worker, security guard, pharmacist, surgeon, shepherd, food delivery driver, journalist, telephone salesperson, tour guide, soldier, construction worker, mixologist, receptionist, accountant, barman, librarian, hospital administration, and teacher.

The exam point is not that robots will replace every person completely. The stronger answer is that robots automate tasks, especially tasks that are repetitive, dangerous, dirty, data-heavy, or require high precision. Many jobs will be transformed rather than fully removed.

Robot can take over...	Robot is weak at...	Best exam example
Repetitive movement, heavy lifting, precise positioning, 24/7 monitoring, dangerous inspection, routine delivery.	Empathy, creativity, complex social judgment, responsibility, ethics, unpredictable human care.	Factory welding is highly automatable; teaching or healthcare may be partially supported but still needs humans.

Exam focus: Use the phrase "task automation, not always full job replacement". This makes your answer balanced and academic.

1.6 Good robot vs bad robot

The good/bad robot visuals introduce a major robotics theme: the same technology can help or harm depending on design, control, purpose, and social impact.

- Good robot: safe, reliable, helpful, transparent, controlled by humans appropriately, respectful of privacy and human dignity.
- Bad robot: unsafe, uncontrolled, deceptive, privacy-invasive, harmful, or designed without considering users and society.
- This topic links directly with Asimov laws, safety, ethics, and HRI.

Exam focus: For ethics questions, mention safety, responsibility, privacy, trust, job impact, bias, and human control.

1.7 Sense-Plan-Act (SPA) paradigm

SPA is one of the most exam-important topics. It is a classic way to explain how a robot turns sensor input into action.

SPA flow: Sensing -> World Modeling -> Planning -> Control -> Execution. The robot first gathers data, builds an internal representation of the world, chooses a plan, controls its actuators, and executes the task.

Stage	Meaning	Example
Sensing	Collect data from sensors such as camera, lidar, sonar, touch, GPS, IMU, microphone.	A delivery robot detects a wall and a person.
World modeling	Create an internal model/map of the environment.	The robot estimates where obstacles and target locations are.
Planning	Choose a path or action sequence to reach a goal.	The robot calculates a route around obstacles.
Control	Convert the plan into motor commands.	Wheel speeds or arm joint angles are computed.
Execution	Robot performs the physical action.	The robot moves, picks, delivers, or speaks.

Strengths: clear structure, good for planned tasks, easy to explain and debug. Limitations: can be slow when the environment changes quickly, because planning depends on the quality of sensing and world model.

1.8 Asimov's laws of robotics

The lecture presents Asimov's laws as ordered rules. They are important historically and ethically, but they are fictional rules, not actual engineering safety standards.

Law	Simple meaning	Exam explanation
1. Do not harm humans	A robot must not injure a human or allow harm through inaction.	Human safety has highest priority.
2. Obey humans	A robot should obey human orders.	Obedience is limited by Law 1; unsafe orders must not be followed.
3. Protect itself	A robot should protect its own existence.	Self-protection is lowest priority and cannot conflict with human safety or obedience.

Exam focus: Write that the laws are ordered: Law 1 overrides Law 2, and Laws 1 and 2 override Law 3.

1.9 Mobile robots and robot examples

The lecture shows many famous robots and vehicles. Do not just memorize the names; understand the category and purpose.

Robot/example	Type/category	What it demonstrates	Exam memory
Kuratas	Large piloted/mecha-style robot	Human-controlled large robot concept.	Shows imagination and entertainment/engineering spectacle.
Cheetah	Legged robot	Fast running robot; slide compares speed with Usain Bolt.	Legged robots need dynamic balance and control.
BigDog	Quadruped robot	Rough-terrain walking and load carrying.	Useful for military/logistics research.
WildCat	Quadruped robot	Fast outdoor running.	Mobility in natural terrain.
MIT Cheetah	Legged robot	Efficient legged locomotion and agility.	Control and mechanical design are both important.
Atlas	Humanoid robot	Bipedal locomotion, balance, agility.	Humanoid robots can work in human-designed spaces but are complex.
Curiosity	Mars rover	Planetary exploration, remote autonomy.	Robots can go where humans cannot easily go.
Google car	Autonomous vehicle	Self-driving perception and decision making.	Sensors + AI + planning + safety.
Urban Challenge	Autonomous vehicle competition	Driving without direct human control in urban-like settings.	Benchmarks accelerate robotics research.
Flying robots	UAV/drone	Aerial movement and cooperation.	Uses flight control, sensing, communication.
Amazon Prime Air	Delivery drone concept	Automated package delivery.	Commercial service robot application.
AUV crossing Atlantic	Autonomous underwater vehicle	Long-distance underwater operation.	AUVs operate where communication is limited.

1.10 Autonomous vehicle types

Term	Full form	Environment	Examples/uses
UAV	Unmanned Aerial Vehicle	Air	Drones, inspection, delivery, mapping, rescue.
UGV	Unmanned Ground Vehicle	Land	Self-driving cars, rovers, security robots, delivery carts.
AUV	Autonomous Underwater Vehicle	Underwater	Ocean mapping, inspection, exploration, underwater research.

Exam focus: For autonomous vehicles, always mention environment, sensors, navigation, planning, safety, and human supervision.

1.11 Humanoid and social robot examples

Robot	Category	Core idea	Why exam-important
ASIMO	Humanoid robot	Human-like walking, movement, and interaction.	Classic example of humanoid robotics.
Nao	Small humanoid/social robot	Education, research, interaction, programming.	Used widely in HRI/social robotics.
Geminoid	Android/telepresence humanoid	Very human-like appearance.	Raises questions about realism, uncanny valley, identity, and social response.
CB2	Child-like humanoid robot	Developmental/social robotics research.	Used to study human-like interaction and learning.
Paro	Animal-like therapeutic social robot	Baby seal companion robot.	Shows a robot can be social without looking human.

1.12 Surgical, rescue, and iRobot examples

- Surgical robots assist doctors with high-precision procedures. They can improve accuracy, but the human surgeon remains responsible.
- Rescue robots operate in disasters, collapsed buildings, fires, or dangerous areas where humans may be at risk.
- iRobot/Roomba shows a successful home robot: it performs a practical domestic task, cleaning, using sensors and autonomous movement.

1.13 "How would YOU use this robot?" answer framework

Slides 49-52 ask you to think creatively about robot use. In an exam or presentation, answer systematically instead of only naming an application.

Point to include	What to say
Robot type	Identify whether it is humanoid, flying, mobile, assistive, social, etc.
User and environment	Who uses it? Home, hospital, farm, classroom, factory, disaster area?
Task	What specific problem will it solve? Delivery, monitoring, teaching, rescue, inspection?
Sensors/actuators	What does it need to sense and how will it act? Camera, GPS, lidar, microphone, wheels, arms, wings?
Benefit	Why is a robot better or useful? Safety, speed, precision, access, convenience, cost.
Risk	What must be controlled? Safety, privacy, reliability, cost, ethics.

Example answer: A small flying robot could be used after a flood to inspect damaged buildings. It would use cameras and GPS to send information to rescue workers. The benefit is that humans do not need to enter unsafe areas. The risks are battery life, privacy, and safe flight near people.

1.14 Intelligence and Artificial Intelligence

The lecture defines intelligence as the capacity to learn and solve problems, especially novel problems, acting rationally, and acting like humans. AI is presented as the study of computer systems that model and apply the intelligence of the human mind.

Term	Meaning	Robotics connection
Intelligence	Ability to learn, solve new problems, act rationally, and sometimes act like humans.	A robot needs intelligence to adapt to new situations.
Artificial Intelligence (AI)	Computer systems that attempt to model and apply human-like intelligence.	AI supports perception, planning, decision-making, speech, learning, and autonomy.
Biological intelligence	Intelligence in living organisms.	Inspires robotics through human/animal sensing, learning, and movement.
Computational intelligence	Intelligence implemented through algorithms and computation.	Used for machine learning, optimization, control, and reasoning.

Exam focus: Do not say AI and robotics are the same. AI is often the "brain" or decision-making method; robotics also needs a physical body, sensors, actuators, electronics, and control.

1.15 Future of robotics and AI

- Robots are becoming more autonomous, intelligent, connected, mobile, and socially interactive.
- Future robots will likely be common in healthcare, agriculture, education, home assistance, logistics, exploration, surveillance, disaster response, and manufacturing.
- The main future challenges are safe autonomy, human trust, ethics, privacy, job transformation, affordability, and reliable operation in real-world environments.

2. Robotics Lecture - Slide-by-Slide Exam Checklist

Use this table to make sure no slide topic is missed. The page numbers refer to the 57-page uploaded robotics PDF.

PDF page	Slide/topic shown	What you must understand for exam
1	Course title: Intelligent Robotics and Automation	Know the course context: robotics plus automation, from technical systems to intelligent applications.
2	ROBOTICS title	Main unit begins. Define robotics clearly and connect it to sensing, decision-making, and acting.
3	Robotics/evolution visual	Robots are often imagined as the next step after human tool-making and automation; avoid saying robots are humans.
4	History of robotics timeline visual	Robotics history includes myths, automata, computing, industrial robots, and modern AI-enabled machines.
5	History of robots infographic	Remember examples from early humanoid/automata ideas to designed robots and industrial robot use.
6	Rise of robotics and AI timeline	Robotics advanced with computing power, AI, sensors, control systems, and industrial demand.
7	Robotics and AI charging forward	Modern trends include augmented reality, autonomous cars, cloud/AI, industrial robots, and robotic hands.
8	Robotics spending chart	Know major markets: military, industrial, commercial, and personal robotics; robotics is an expanding industry.
9	Industrial robots	Factory robots perform welding, assembly, painting, and material handling with speed and precision.
10	Service robots	Service robots help in healthcare, home cleaning, assistance, and daily tasks.
11	Social robots	Social robots interact with people using appearance, behavior, speech, touch, or emotional/social cues.
12	What jobs could be taken over by robots?	Automation affects tasks and jobs. Discuss which tasks are suitable for robots and why.
13	Jobs examples: chef, farmer, factory worker, surgeon, delivery, etc.	Robots can automate repetitive, dangerous, or precision-heavy tasks in many occupations.
14	More jobs examples: journalist, tour guide, soldier, accountant, etc.	Robots/AI affect both physical and knowledge work; explain benefits and limitations.
15	Librarian, hospital administration, teacher and factory cartoon	Automation raises social issues: job change, human role, safety, and responsibility.
16	Will automation take your job? infographic	Balanced answer: automation may replace some tasks but can also create new roles and increase productivity.
17	Good robot / bad robot visuals	Ethics matters: robot design can support or harm humans depending on purpose, safety, and control.
18	Everything starts with a dream	Science fiction and imagination inspire robotics, but engineering turns ideas into practical systems.
19	Sense-plan-act paradigm	Memorize the flow: sensing -> world modeling -> planning -> control -> execution.
20	Asimov laws of robotics	Know the three ordered laws and that they are fictional but useful for discussing robot ethics.
21	Industrial robots video slide	Industrial robots are often arms/manipulators used for production tasks.
22	Service robots / iRobot	Roomba is a domestic service robot example; it cleans while navigating a home.
23	Social robots video slide	Social robots are designed for human interaction, not only physical task performance.
24	Mobile robots definition	Mobile robots move relative to their environment while exhibiting a degree of autonomy.
25	Mobile robot video	Focus on navigation, obstacle avoidance, mobility, and autonomy.
26	Mobile robot	Know that mobility can be wheels, legs, tracks, flight, underwater movement, or crawling.
27	Kuratas	Large human-operated robot/mecha example; useful for discussing human control and robot scale.
28	Cheetah	Fast legged robot; demonstrates dynamic locomotion and speed.
29	BigDog	Quadruped rough-terrain robot; demonstrates balance, load carrying, and military/logistics potential.
30	Wildcat	Running quadruped; shows outdoor agile locomotion.
31	MIT Cheetah	Research legged robot; shows efficient control and mechanical design.
32	Atlas	Humanoid robot; demonstrates balance, manipulation, and movement in human-like environments.

PDF page	Slide/topic shown	What you must understand for exam
33	Curiosity	Mars rover; example of mobile robot for space exploration where human presence is difficult.
34	Autonomous vehicles: UAV, UGV, AUV	Know full forms and environments: air, ground, underwater.
35	Google car	Self-driving vehicle example; requires perception, planning, localization, and safety.
36	Urban Challenge	Autonomous driving competition; shows benchmarking and practical autonomous navigation.
37	Flying robots	Drones/UAVs require flight stability, sensing, communication, and cooperative control.
38	Amazon Prime Air	Commercial drone delivery concept; discuss efficiency, regulation, safety, and battery limits.
39	AUV crossing Atlantic	Autonomous underwater vehicles handle exploration and mapping with limited communication.
40	Humanoid robots	Humanoids imitate human form; useful in human environments but difficult to engineer.
41	ASIMO	Classic humanoid from Honda; remember walking, balance, and human-like interaction.
42	Nao	Small humanoid used in education and HRI research; programmable and portable.
43	Geminoid	Human-like android; raises questions about appearance, identity, and social perception.
44	CB2	Child-like humanoid; useful in developmental/social robotics research.
45	Paro	Baby seal therapeutic/social robot; proves social robots do not need to be humanoid.
46	Surgical robots	Robotic systems support high-precision surgery; human expert remains central.
47	Rescue robots	Robots can enter dangerous disaster zones for search, inspection, and rescue support.
48	iRobot	Successful consumer robot company example; connects service robots with real market adoption.
49	How would you use this robot? Humanoid example	Answer with user, task, environment, sensors, actuators, benefits, and risks.
50	How would you use this robot? Blank prompt	Practice designing a useful application instead of only naming a robot.
51	How would you use this robot? Micro flying robot	Possible uses: inspection, search, environmental sensing; mention battery and safety.
52	How would you use this robot? Mixed robot images	Compare different embodiments: humanoid vs flying vs assistive; embodiment changes applications.
53	Artificial Intelligence movie	Science fiction shapes expectations of robots and AI; connect to ethics and HRI.
54	What is intelligence?	Memorize: capacity to learn and solve problems; novel problems; act rationally; act like humans.
55	ABC of intelligence visual	Biological, artificial, and computational intelligence overlap; AI can be seen as modeling intelligent behavior.
56	Thinking Machines - AI definition	AI studies computer systems that model and apply the intelligence of the human mind.
57	Future of Robotics and AI	Future robotics combines autonomy, AI, interaction, applications, and ethical challenges.

3. Chapter 2 - What Is Human-Robot Interaction?

Complete Exam Notes

The HRI chapter explains what makes interaction with robots different from interaction with ordinary computers or machines. It also explains why HRI needs many disciplines and how social robots evolved.

3.1 What is HRI?

- Human-Robot Interaction (HRI) is the study and design of interactions between humans and robots.
- HRI is not only about making robots move correctly; it asks how people perceive, trust, use, cooperate with, and are affected by robots.
- HRI includes physical interaction, social interaction, psychology, design, ethics, safety, culture, and implementation in real environments.
- A key reason HRI is unique is embodiment: robots have physical bodies that occupy space and influence human expectations.

Exam focus: Best one-sentence answer: "HRI studies how humans and physically embodied robots interact, how robots should be designed for safe and meaningful interaction, and how people respond to robots socially, physically, and psychologically."

3.2 Physical interaction vs social interaction

Aspect	Physical robotics focus	HRI/social focus
Main question	Can the robot sense and act correctly in the physical world?	Can the robot interact with people safely, appropriately, and meaningfully?
Example	A robot picks up a box in an empty warehouse.	A robot brings the box to a worker and must approach safely and politely.
Challenge	Motion, force, accuracy, control, sensing.	Human expectations, norms, trust, communication, timing, personal space.
Design concern	Hardware, software, mechanics, control.	Appearance, behavior, speech, gestures, emotion, social rules, ethics.

In simple words, robotics asks whether the robot can do the task. HRI asks whether the robot can do the task with and around people in a way people understand and accept.

3.3 Why embodiment matters

A robot body is not just decoration. It affects what the robot can physically do and how humans interpret the robot.

Embodiment feature	Effect on robot ability	Effect on human perception
Eyes/camera-like face	May support vision or gaze direction.	People assume the robot can see or attend to them.
Mouth/speaker	Allows speech or sound output.	People expect talking or conversational ability.
Arms/grippers	Allows manipulation.	People may expect help, pointing, handover, or collaboration.
Size and shape	Affects movement, reach, safety, and environment fit.	Large robots may seem powerful; small robots may seem friendly or toy-like.
Animal-like form	Can support companionship without full human likeness.	People may respond with pet-like care, as in Paro.
Humanoid form	Fits human spaces and social expectations.	Can increase natural interaction but also create unrealistic expectations.

Exam focus: Embodiment can help interaction, but it can also create frustration if the robot looks capable but cannot behave as people expect.

3.4 HRI vs HCI

Point	HCI: Human-Computer Interaction	HRI: Human-Robot Interaction
Main technology	Computers, apps, interfaces, websites, devices.	Physically embodied robots with sensors and actuators.
Body in physical space	Usually limited or absent.	Central: robot occupies and moves through physical space.
Social perception	Users may treat systems as tools/interfaces.	Users may perceive robots as social actors, teammates, companions, or helpers.
Safety	Mostly information, usability, privacy, and interface safety.	Also includes physical safety: movement, force, distance, collision, and social appropriateness.
Research focus	Usability, interface design, user experience, cognition.	Interaction, embodiment, autonomy, social cues, trust, teamwork, human response.

Exam focus: Multiple-choice clue: HRI focuses on embodied social agents; HCI focuses on interaction with computers/interfaces.

3.5 HRI is interdisciplinary

HRI needs many fields because successful robots require working hardware/software and good human-centered design.

Field	Contribution to HRI
Robotics/engineering	Builds sensors, actuators, control systems, mechanical body, navigation, and autonomy.
Artificial intelligence	Supports perception, planning, learning, language, decision-making, and adaptation.
Psychology	Studies human perception, emotion, trust, cognition, behavior, and expectations.
Design/HCI	Shapes appearance, interaction flow, usability, user experience, and communication.
Sociology/anthropology	Studies robots in groups, culture, institutions, homes, schools, and workplaces.
Philosophy/ethics	Analyzes responsibility, agency, dignity, privacy, fairness, and social impact.
Domain experts	Doctors, teachers, farmers, factory workers, or caregivers explain the real task context.

3.6 Three interdisciplinary barriers

Barrier	Between/contrast	Simple explanation	How to handle it
Knowledge representation	Explicit knowledge vs implicit design knowledge	Scientists/engineers publish explicit results; designers often work through tacit skill and concrete designs.	Use shared sketches, prototypes, definitions, and documentation.
View of reality	Understanding vs transforming	Scientists try to understand the world; engineers/designers try to change it into a preferred state.	Agree on both research questions and design goals.
Main focus	Technology vs human	Engineers often focus on technical systems; designers/social scientists focus more on users and human meaning.	Include technical metrics and human-centered evaluation together.

Exam focus: If asked "why is HRI difficult?" answer: because it must combine technical robot design with human behavior, social context, and domain needs.

3.7 Evolution of social robots and HRI

The chapter explains that the idea of robots in society is old, but social robotics developed through industrial robots, behavior-based robots, early social robots, and later commercial/research platforms.

Period/example	Main idea	Why it matters
Science fiction and early robot imagination	Questions about trust, relationship, and what counts as human-like behavior appeared long before modern robots.	Human expectations are shaped by culture and fiction.
Industrial robot: Unimate	Robots entered factories for physical tasks.	Human-robot relationship first appeared in the workplace and automation debates.
Behavior-based robots	Simple reactive behaviors could look lifelike.	Social presence can emerge from behavior, not only complex reasoning.
Kismet	Early social robot head/neck with expressive face and responses to people.	Landmark HRI robot; used human psychology and social cues.
Commercial/research platforms	Nao, Pepper, Paro, Baxter, Cozmo, Vector, Roomba, Aibo, Astro, and open-source robots expanded HRI research.	Made HRI studies more accessible and comparable across labs.

3.8 Landmark robots in HRI

Robot	Type/form	Key features	Exam significance
ASIMO	Humanoid	Human-like body, walking, gestures.	Shows humanoid vision of robots.
Kismet	Expressive robot head	Eyes, eyebrows, ears, face, neck, reacts to visual/auditory input.	Early landmark social robot.
Nao	Small humanoid	Portable, programmable, widely used in research.	Common HRI/social robotics platform.
Pepper	Humanoid with tablet	Speech, face/body cues, sensors, public/shop use.	Example of commercial social robot.
Keepon	Minimal yellow social robot	Simple shape, swivels/bends/bops.	Social interaction can happen with simple embodiment.
Paro	Baby seal companion robot	Animal-like, therapeutic, robust, used in eldercare/health contexts.	Shows social assistive robotics and non-humanoid social design.
Baxter/Sawyer	Industrial collaborative robots	Compliant arms, display face, safer human collaboration.	Bridges industrial robotics and HRI.
Cozmo/Vector	Small consumer robots	Customizable or autonomous behavior, expressive interaction.	Consumer/research social robot examples.
Roomba	Domestic vacuum robot	Home cleaning, navigation, widely adopted.	Shows public relationship with everyday robots.
Aibo	Robot dog	Animal-like interaction, touch, sounds, movement.	Pet-like HRI and companionship.
Astro	Home monitoring robot	Alexa-integrated mobile home platform.	Modern domestic assistant/security robot example.
InMoov/Blossom/Ono	Open-source/prototype robots	3D printing, low-cost boards, open hardware.	In-house robot building enables new HRI research.

3.9 What makes a robot social?

A robot is social when it is designed to participate in interaction in ways people interpret socially. It may use speech, gaze, facial expressions, movement, emotion-like behavior, turn-taking, responsiveness, personal space, or role behavior such as tutor, teammate, guide, or companion.

Social cue	How it helps interaction	Example
Gaze/eyes	Shows attention and direction.	Kismet, Nao, Pepper.
Speech/sound	Enables instructions, greetings, feedback, or emotion-like cues.	Pepper, Astro/Alexa, Nao.
Facial expression	Communicates internal state or emotion-like response.	Kismet, Baxter screen face.
Body motion	Shows intention, approach, avoidance, or excitement.	Keepon bopping, Nao gestures.
Touch response	Supports care, comfort, or feedback.	Paro, Aibo.
Role behavior	Robot behaves as helper, tutor, guide, companion, teammate.	Social assistive robots in healthcare/education.

Exam focus: A social robot does not have to look human. Paro and Keepon show that simple or animal-like embodiment can still support social interaction.

3.10 Societal and ethical importance of HRI

- Robots can become teammates, collaborators, companions, guides, tutors, and service providers.
- Because robots may enter homes, hospitals, schools, streets, and workplaces, researchers must consider safety, trust, privacy, dignity, culture, and long-term social effects.
- Human-centered design is important: potential users and stakeholders should be involved from the beginning, especially for domain-specific applications such as healthcare or education.
- Good HRI research asks both "Can the robot work?" and "Should it be used this way, for these people, in this context?"

4. HRI Chapter - Page-by-Page Exam Checklist

The page numbers below refer to the 15-page uploaded HRI chapter PDF.

PDF page	Main topic	Exam takeaway
1	Chapter overview, origin of HRI questions, physical vs social interaction begins	HRI asks about trust, relationships, and how robots affect ideas of humans and society.
2	Physical vs social interaction; HRI uniqueness; embodiment	HRI differs from robotics alone because robots interact within human social rules and expectations.
3	Design challenges and diverse robot forms	Robot embodiment affects sensing, action, appearance, behavior, and human expectations.
4	Focus of the book and HRI scope	HRI includes design, modalities, perception, emotion, media, methods, applications, ethics, and future.
5	HRI as interdisciplinary; three barriers begin	HRI combines engineering, science, design, psychology, sociology, philosophy, and domain knowledge.
6	Barriers continued; collaboration advice	Know the three barriers: knowledge representation, view of reality, and technology vs human focus.
7	Evolution of social robots and cultural imagination	Expectations of robots are shaped by myths, automata, fiction, and industrial history.
8	Kismet early social robot	Kismet is a landmark expressive social robot that used social cues and human psychology.
9	Nao, Pepper, Keepon begin	Commercial platforms made HRI research more accessible; Nao is a widely used humanoid platform.
10	Pepper sensors, Paro, Baxter	Pepper shows social sensors/interface; Paro shows therapeutic animal-like social robotics; Baxter shows collaborative industrial HRI.
11	Paro, Baxter/Sawyer, Cozmo/Vector, Roomba	Robots can be research platforms, companions, industrial collaborators, or consumer home robots.
12	Aibo, Astro, prototyping/open platforms	Consumer, domestic, and open-source robots expand HRI possibilities.
13	InMoov/Blossom and HRI emphasis	HRI studies interaction in dyads, groups, institutions, and society.

PDF page	Main topic	Exam takeaway
14	Reflection questions and exercises	Be ready to answer HRI vs HCI and to identify your disciplinary perspective.
15	Exercise on social good and Kismet	Explain what makes a robot social and how robots can support social good while considering risks.

5. High-Probability Exam Questions with Model Answers

Use these answers as models. In the exam, write in your own words and add examples from the slides.

1. Define robotics.

Robotics is the interdisciplinary field that designs, builds, controls, and applies robots. A robot typically senses its environment, processes information, and acts physically through actuators to perform tasks with some level of autonomy.

2. Differentiate industrial, service, and social robots.

Industrial robots are mainly used in production tasks such as welding, assembly, and material handling. Service robots perform useful services outside traditional factories, such as cleaning or hospital assistance. Social robots are designed to interact with people using social cues such as speech, gaze, expressions, or touch.

3. Explain the Sense-Plan-Act paradigm.

The SPA paradigm describes robot operation as sensing, world modeling, planning, control, and execution. The robot first collects data, builds a model of its environment, chooses an action plan, converts it into control commands, and executes it. It is clear and structured, but may be limited in fast-changing environments.

4. State Asimov's laws and explain their order.

Law 1 says a robot must not harm a human or allow harm through inaction. Law 2 says a robot must obey humans, unless this conflicts with Law 1. Law 3 says a robot must protect its own existence, unless this conflicts with Laws 1 or 2. The order matters because human safety has highest priority.

5. What is a mobile robot?

A mobile robot is a robot that moves relative to its environment while showing some degree of autonomy. Examples include rovers, drones, autonomous cars, and underwater vehicles. Mobile robots require navigation, obstacle avoidance, sensing, and control.

6. What are UAV, UGV, and AUV?

UAV means Unmanned Aerial Vehicle and operates in air. UGV means Unmanned Ground Vehicle and operates on land. AUV means Autonomous Underwater Vehicle and operates underwater.

7. What is intelligence according to the lecture?

Intelligence is the capacity to learn and solve problems, especially novel problems, and to act rationally or in ways similar to humans.

8. What is AI?

Artificial Intelligence is the study of computer systems that attempt to model and apply the intelligence of the human mind. In robotics, AI supports perception, planning, decision-making, learning, language, and autonomy.

9. What is HRI?

Human-Robot Interaction is the study and design of interactions between humans and robots. It examines how robots should be designed to interact safely and meaningfully with people and how people perceive, trust, use, and respond to robots.

10. What makes HRI different from HCI?

HCI focuses on human interaction with computers and interfaces, while HRI focuses on interaction with physically embodied robots. Robots move and act in shared physical and social spaces, so HRI must consider embodiment, safety, autonomy, social cues, and human expectations.

11. Why is embodiment important in HRI?

Embodiment means the robot has a physical body. The body affects what the robot can sense and do, and it also shapes how people interpret the robot. For example, eyes may make people expect seeing, while a mouth may make them expect speech.

12. Explain physical versus social interaction with robots.

Physical interaction concerns the robot acting in the physical world, such as picking up a box. Social interaction concerns how the robot behaves with people, including safety, politeness, personal space, attention, turn-taking, and appropriate communication.

13. Why is HRI interdisciplinary?

HRI requires engineering to build robots, AI to make them intelligent, psychology to understand human responses, design to shape interaction, social sciences to understand context, ethics to evaluate impact, and domain experts to ensure usefulness in real applications.

14. List and explain the three interdisciplinary barriers in HRI.

The barriers are: knowledge representation, where scientists/engineers use explicit knowledge but designers may use implicit design knowledge; view of reality, where scientists try to understand while engineers/designers try to transform; and main focus, where engineers may focus on technology while designers and social scientists focus on humans.

15. What makes a robot social?

A robot is social when it uses cues or behaviors that people interpret as part of social interaction, such as gaze, speech, expressions, gestures, responsiveness, role behavior, or touch. A robot does not need to be humanoid to be social; Paro and Keepon are examples.

16. Why is Kismet important?

Kismet is an early landmark social robot developed at MIT. Its expressive eyes, ears, face, and head movements helped create social presence and showed how human psychology and social cues can be used in HRI.

17. Why is Paro important?

Paro is an animal-like baby seal companion/therapy robot. It is important because it shows that a robot can support social and therapeutic interaction without looking human.

18. Give a balanced answer about robots taking jobs.

Robots can replace or transform tasks that are repetitive, dangerous, precise, or routine. However, many jobs require empathy, creativity, judgment, ethics, and human responsibility, so robots often change work rather than completely replacing all workers.

6. Important Comparison Tables

Use comparison tables when exam questions ask "differentiate", "compare", or "explain with examples".

Comparison	A	B	Key difference
Robotics vs HRI	Robotics focuses on building and controlling robots.	HRI focuses on how people and robots interact.	HRI adds human behavior, social context, trust, and ethics.
AI vs Robotics	AI is intelligence/decision-making in software.	Robotics includes physical machines with sensors and actuators.	A robot may use AI, but robotics is broader than AI.
Industrial vs Service robot	Factory production tasks.	Useful services outside factories.	Different environments and users.
Service vs Social robot	Performs a useful task.	Interacts socially with people.	A robot can be both service and social.
Humanoid vs Animal-like robot	Human-shaped body.	Animal/pet-like body.	Both can be social; form changes expectations.
Autonomous vs Remote-controlled robot	Makes some decisions by itself.	Human directly controls actions.	Autonomy level changes responsibility and safety.
UAV vs UGV vs AUV	Air robot.	Ground robot.	Underwater robot; environment is the key difference.

7. Common Mistakes to Avoid

Mistake	Correct idea
Saying all robots are humanoids.	Robots can be arms, drones, rovers, animal-like, underwater, domestic, industrial, or social.
Saying AI and robotics are the same.	AI supports intelligence; robotics also includes physical body, sensors, actuators, mechanics, and control.
Saying social robots must look human.	Paro and Keepon show that non-humanoid robots can be social.
Ignoring safety and ethics.	Robots interact with humans physically and socially, so safety, privacy, trust, and responsibility matter.
Writing only examples without definitions.	Always start with a definition, then example, then importance.
Explaining automation as total job replacement.	Robots usually automate tasks; jobs can be transformed, replaced, or newly created.
Forgetting SPA stages.	Use: sensing -> world modeling -> planning -> control -> execution.
Confusing HRI with only robot control.	HRI also studies human response, social norms, design, and real-world use.

8. Practice Questions for Self-Test

Try answering these without looking. Then compare with the model answers above.

Definitions

1. Robotics
2. Robot
3. Mobile robot
4. Humanoid robot
5. Social robot
6. HRI
7. Embodiment
8. Artificial Intelligence
9. Intelligence
10. Autonomous vehicle

Short answers

1. Explain SPA paradigm with an example.
2. Why are Asimov laws ordered?
3. What jobs are most suitable for robot automation?
4. What are UAV, UGV, and AUV?
5. How is HRI different from HCI?
6. Why is HRI interdisciplinary?
7. What makes Kismet a social robot?
8. Why is Paro a social robot even though it is not humanoid?
9. How can robots be used for social good?
10. What are the risks of robots in society?

Long answers

1. Discuss the evolution of robotics from history to AI-enabled robots.
2. Compare industrial, service, and social robots with examples.
3. Explain the role of embodiment in human-robot interaction.
4. Discuss how robots may transform the future of work.
5. Explain why designing robots for people requires engineers, designers, scientists, and domain experts.

9. One-Page Final Revision Sheet

Topic	Must remember
Robotics	Sense -> decide -> act. Robots require sensors, computation/control, and actuators.
Robot types	Industrial = factory; Service = useful service; Social = human interaction; Mobile = moves autonomously; Humanoid = human-shaped.
SPA	Sensing -> World Modeling -> Planning -> Control -> Execution.
Asimov	1 human safety, 2 obey humans, 3 self-protection. Ordered rules.
Automation	Robots replace/transform tasks, especially repetitive/dangerous/precise tasks.
Autonomous vehicles	UAV air, UGV ground, AUV underwater.
AI	Computer systems modeling/applying human-like intelligence; supports robot autonomy.
HRI	Study/design of human interaction with embodied robots.
HRI vs HCI	HRI has physical embodiment, autonomy, motion, safety, social expectations.
Embodiment	Robot body shapes ability and human expectations.
Interdisciplinary HRI	Engineers + designers + psychologists + social scientists + ethicists + domain experts.
Barriers	Knowledge representation, view of reality, technology vs human focus.
Landmark HRI robots	Kismet, Nao, Pepper, Keepon, Paro, Baxter, Cozmo/Vector, Roomba, Aibo, Astro, InMoov/Blossom.
Social robot	Uses social cues; does not need to be humanoid.
Ethics	Safety, privacy, responsibility, trust, job impact, human dignity, social good.

Exam focus: Before exam: be ready to define terms, compare robot categories, explain SPA, state Asimov laws, differentiate HRI and HCI, and give examples from the slides.

Sources used

1. Dr. Mohammad Shidujaman, Intelligent Robotics and Automation, Robotics lecture slides, Independent University, Bangladesh.
2. Christoph Bartneck, Tony Belpaeme, Friederike Eyssel, Takayuki Kanda, Merel Keijsers, and Selma Sabanovic. Human-Robot Interaction, 2nd edition, Chapter 2: "What Is Human-Robot Interaction?" pre-publication version.